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SECTION 1

Modernism and its Discontents

Meaning and Effect: Revisiting Semiotics in Architecture

André Loeckx and Hilde Heynen

Semiotics 1960-1980: A new figure of knowledge for architectural theory

Semiotics did not originate in the mid-1960s. At the awakening of the 20th century, the development of a general “theory of signs” was announced almost simultaneously by the Swiss linguist Ferdinand de Saussure (1857-1913), who named it “sémiologie,” and by the American philosopher and scientist Charles Sanders Peirce (1839-1914), who used the term “semiotic.”¹ In Continental Europe Saussure’s ideas matured slowly within language studies. In the Anglo-American context, Peirce’s theory of signs survived under the wings of philosophical Pragmatism and behavioral sciences. These two semiotic strands differed from one another, in that Saussure’s theory focused on the signification of signs, whereas Peirce’s thinking gave preference to the working (“semiosis”) and the effect of signs. Both strands developed rather independently from one another, although there were some crossovers, as we will see.

From the late fifties onwards, the widespread success of structuralism, which was informed by Saussurean structural linguistics, generated a spectacular resurrection of semiotics, not just in linguistics but also in social sciences and in the arts, which suddenly began to see their own object of study as structured just like languages. This “linguistic turn” also influenced the field of architecture. Indeed, many architectural scholars turned to semiotics in the 1960s and 1970s, in response not just to the overall surge of structuralism but also as a possible way out of the widely perceived crisis of legitimacy of modern architecture. Architectural historian and theorist Françoise Choay (1925) thought for example that investigating semiotics could help understand the process of “semantic reduction” that affected so many postwar housing estates, which displayed what she called a “hyposignifiante” of the daily life environment.² In such a process the “hypersignifiante,” that is, the saturated meaning of other, more traditional environments (Choay referred to the Bororo village, the ancient Greek agora, and medieval Fribourg) was gradually replaced

by one singular stratum of meaning in the modern city: the economic. Semiotics seemed to offer a way out of this impoverishment by providing an understanding of the conditions for contemporary signification in architecture and urbanism.

In hindsight, one can see that the turn to semiotics in architecture was remarkably short-lived, with virtually all the most relevant texts published between 1960 and 1980. It nevertheless included some attempts for a critical engagement with architecture that are worth revisiting. In this approach architecture is seen as a domain of general semiotics, or put more specifically, as a kind of language. Viewing architecture as language was not a new phenomenon: it had never been entirely absent in architectural thought. This is particularly true for the tradition of classical architecture, which tended to conceive of architecture as an autonomous visual language determined by an underlying system of rules.³ Several architects and theorists of the Modern Movement, however, turned against the classical conception of architecture as a language that serves symbolic representation. The idea that architecture could represent something (status, majesty, power) outside its function was considered a fallacy. The “struggle of modern architecture,” according to Mart Stam, was “a struggle against the representative, against excess.”⁴ Representation, in this view, was seen as corroding “the truth of the form” and encouraging the formal lie of style imitation and decoration.⁵ Louis Sullivan’s dictum “form follows function,” Adolf Loos’s famous tirade against the decadence of the ornament, and Le Corbusier’s purism are all instances of modern architecture’s advocacy of a mode of design that naturally emerges from an inner truth of logical objectives and functionality and that would thus comply with the rationality of nature, science, economics, and culture.⁶ Form was seen as the outward figuration of that inner truth; furthermore, symbolic references to dubious external factors were suppressed.

In an important text from 1957, John Summerson considered the rupture with the classical language system an essential characteristic of a modern architecture that adopted the program as the source of the unity of a design. Nonetheless, “the conceptions which arise from a preoccupation with the programme have got, at some point, to crystallise into a final form,” whereas “there is no common theoretical agreement as to what happens or should happen at that point. There is a hiatus. One may even be justified in speaking of a *‘missing architectural language’* [our italics].”⁷ According to Summerson, this generated a need for a more elaborate and self-reflective architectural theory.

Summerson’s diagnosis was symptomatic of the confusion among architects as to how form and content, as well as aesthetics and utility, should relate to each other. They turned to semiotics because of its apparent promise to clarify the troubling relationship between form on the one hand and “content” or “meaning” on the other, a relationship considered similar to the semiotic definition of the sign.

Semiotics was expected to provide a scientifically operational definition of “the sign,” which could be transferred to architecture. Unfortunately, all over the semiotic field, this definition generated some serious scholarly divergences. In fact, this field had become a dense “forest of symbols,” characterized by controversial issues, fascinating crossovers, and meandering paradigms.⁸

Our aim in this chapter is to offer a mapping of this “forest of symbols” and of its effects on architectural theory. Unlike Louis Martin’s earlier discussion of architectural semiotics, which deals with the Anglo-American discourse until 1976,⁹ we do not limit ourselves to this brief period and this one paradigm. Our contribution also focuses on European semiology and includes the shift from structuralism to post-structuralism. Its aim is to critically reread a selection of relevant episodes, discussing both the paths taken as well as the opportunities missed with the ultimate intention of rethinking their merits and of reassessing whether some aspects of these semiotic adventures might not deserve to be redeemed rather than discredited.

Semiotic paradigms and passages to architecture

French semiology and the language of architecture

The contribution to structural linguistics by Ferdinand de Saussure situates itself at the level of *langue* (that is, the systematic or structural level of language), rather than dealing with *parole* (the use of language in context).¹⁰ Within this system, Saussure’s definition of the sign is based on two signifying operations that occur simultaneously and that activate two forms of relationality: a positive *association* and a negative *differentiation*. The sign is the *association* of “the signifier” (*le signifiant*), which is usually an “acoustic image,” and “the signified” (*le signifié*), defined as a content or a concept, which does not necessarily refer to “something” out there in the world. According to Saussure, this association is arbitrary and unmotivated. It is, as it were, an alliance sealed by social convention. Within the signifier there is nothing specific that would refer to the signified: the acoustic images “tree” and “*arbre*” are totally different but refer to the same concept. So while the arbitrariness of the association affirms the difference between signifier and signified, the sealed character of the association negates that difference. Saussure however also understood language as a system held together by structural *differentiation*, which is connected to the sign’s position within the overall system or, put otherwise, its “value” within a structure. In short, the sign acquires meaning in the simultaneous play of association and difference. Literary theorist Roland Barthes (1915-1980) clarified this twofold definition with a metaphor borrowed from Saussure himself: signs are like the snippets or shreds that are left after the cutting up of a piece of paper.¹¹ Every snippet has a recto and

a verso (a signifier and a signified) and, at the same time, has its value fixed by its relative position in relation to the surrounding snippets.

From Saussure's twofold definition (association and differentiation) follows that the system of signs – language – does not merely serve to represent an outside reality or to display a number of preexisting meanings. Rather, meaning and language originate in the simultaneous articulation of what can be heard and what can be thought. In fact, this perspective asserts that the only cognizable, meaning-bearing reality is a linguistic reality. The thin materiality of the Saussurean sign, the arbitrary character of the association between signifier and signified and the absence of “a reality out there” to which the sign refers raised many objections but would also generate innovative amendments that later on would fuel post-structuralist thinking. These issues would also continue to differentiate Continental-European semiology from the Anglo-American semiotic paradigm.

Another important pair of notions in the Continental-European tradition is that of “syntagm” versus “paradigm,” as explained by Barthes. “Syntagm” refers to the methodical stringing together of signifying elements on the basis of syntactic rules, and this process is related to metonymy (a figure of speech that replaces the name of a thing with the name of something else which is closely related). “Paradigm,” on the other hand, refers to the selection of a signifying term from an associative series of similar terms and has to do with metaphor (a figure of speech that makes an implicit, implied, or hidden comparison between two things that are unrelated but which share some common characteristics). These notions make it possible, according to Barthes, to extend structural linguistics towards a structural semiology of fields such as fashion, food, furniture, or architecture, which can be seen as nonverbal and very material systems of signification.¹² Barthes thus corroborated Saussure's earlier statement that all domains of culture, science, and society in fact could be considered as various forms of language, and that hence his linguistics in the long run might become just another part of a more general science of semiology that would deal with all kinds of languages. Interestingly, Saussure himself explained syntagmatic stringing together and paradigmatic selecting by way of an architectural metaphor. He described how a column is “syntagmatically” connected with other parts of the construction (for example, the cornice), whereas the column itself is selected from a paradigmatic series of Doric, Ionic, or Corinthian versions. This was a very early indication that something like an architectural semiotics might be possible.¹³

Barthes's brilliant collection, *Mythologies* (1957), seemed to open a pathway towards a semiology of nonverbal languages. The book deals with a variety of topics, such as the Tour de France, publicity posters, catch-competitions, or even the avant-garde design of the Citroen DS, considering all of them as languages. Barthes analyzes their mythological effects as resulting from the interplay between “denotation” (a straightforward, literal meaning based on a simple and direct relation between a clear signifier

and an obvious signified) and “connotation” (a more layered and implicit meaning beyond the literal one).¹⁴ According to him, this interplay allows for the insertion of ideology in the language of sport, publicity, or design, because the apparently innocent denotation functions as a vehicle for ideologically charged connotations.

Whereas *Mythologies* offered a very promising start for a semiology of nonverbal systems, Barthes did not continue in the same vein. His *Système de la Mode* (1967), for instance, opts for a semiotic analysis of texts *about* fashion, rather than focusing on fashion itself. The issue of the nonverbal systems of signification, however, continued to intrigue Barthes. In an article that same year, “Sémiologie et Urbanisme,” he claimed that urban spaces constitute in a certain way a discourse in themselves, because they always already act as signifiers. A really scientific semiotics of the city would imply, however, that one has to transition from the simplistic use of the metaphor of “the city as language” towards a much more methodical description and analysis of this “language of the city” (rather than analyzing the language about the city).¹⁵

Anglo-Saxon foundations of a semiotic pragmatism

Charles Sanders Peirce (1839-1914) and Charles W. Morris (1901-1979) are considered the foundational thinkers of a “semiotic” science, the Anglo-American counterpart to Saussure’s semiology, both terms derived from the Greek “semeion” (sign). Peirce’s definition of the sign is triadic: the sign consists of a “representamen” (an object, a characteristic, a thought, a pronouncement, an action...) that refers to an object (that can be a real one); it does so in such a way that this reference evokes a possible “interpretant,” i.e., not an interpreter but an interpretative result, which could include information, a thought, an insight, a reaction, a new reference, or something else. Peirce’s definition is quite different from the Saussurean one.¹⁶ Whereas Saussure considers the sign as part of a language system irrespective of the world outside of that system, Peirce focuses on processes of signification (or “semi-osis”) that can refer to or have an effect in the physical world. Peirce’s definition of the sign is moreover more inclusive than Saussure’s, because it explicitly posits many different things, objects, gestures, concepts, or characteristics as potential representamens. This means that Peirce’s conceptualization facilitates the application of his apparatus to architecture and its materiality.

A crucial contribution of Peirce’s semiotics concerns his ideas about “the three modes of being.” “Firstness” – the mode of being related to potentiality – pertains to what makes the representamen present (a color, a shape, a movement, a sound...). Based on this potentiality, “secondness” – the mode of being of the factual – can operate, whereby the representamen refers to an object that can be “out

there” in the world. This referring process is conditioned by that of having a certain effect – “thirdness” – which is actualized or at least expected. Thirdness hence is the mode of being that has to do with a result, an effect, reaction, or interpretation that can be understood on the basis of a rule, a code, or a convention.¹⁷ The three best-known classes of signs that Peirce discusses are presented in this context, when he deals with the impact of the three modes of being on the factual reference of the representamen to the object. When this reference mainly activates a potentiality (in cases where something resembles something else, like a roof resembling billowing sails), it has to do with an “icon.” When the reference is based on a factual, functional, or causal relationship, the sign is an “index” (such as in the case of a shell roof construction). The “symbol,” lastly, is the class of signs where the reference is understandable because a certain code or convention is applicable (the case of a well-known and much debated modern building).¹⁸

Peirce is one of the founding fathers of philosophical Pragmatism, which understands knowledge of the world as inseparable from agency within it.¹⁹ This is also characteristic of the semiosis of the Peircean sign, which is motivated by the expectation of an effect (the interpretant or interpreting result).²⁰ Often the result or effect is provisional, evoking a new reference that in turn generates another reference. It is possible to compare Peirce’s understanding of semiosis to the endless chain of signifiers that Derrida detects in any given text or reality (the “traces” that always lead to other traces without ever ending at the discovery of “truth”).²¹ For Peirce, however, “unlimited semiosis” does not have the same consequences as for Derrida, for whom meanings are always already provisional from the start, taking part in an endless chain, and thus negating the possibility of an ultimate, transcendental meaning (such as God). In contrast, Peirce’s “pragmatic maxim” prevents unlimited or undetermined semiosis.²² This means that the process of semiosis progresses until a point where “the significance” can be determined as the entirety of effects and interpreting thoughts evoked by the representamen.²³

Equally pragmatic is Peirce’s understanding that the dosage of three “modes of being” activated in the sign – firstness, secondness, and thirdness – is driven by the expected result or effect. Thus, the usage of signs relies on an anticipative, “what if” mode of reasoning that Peirce labels “abduction” – complementing the better known modes of induction and deduction.²⁴ Given the role of abduction in Peirce’s theory of the sign, it is not difficult to see its appeal to architects, whose designerly way of thinking is equally based on an anticipative, “what if” reasoning.

Morris – also an important Pragmatist – likewise offered interesting ideas for an empirically relevant semiotics. In his early writings, Morris defined, just like Peirce, the sign as a triadic relation between a sign vehicle, an object, and an interpretant.²⁵ Later on, he complicated the definition of the sign by describing semiosis as a relation between five poles: a sign vehicle (A) evokes in an interpretor or

addressee (B) a disposition (the interpretant, C) to react in a certain way to the object (D) and this under certain conditions (E). For Morris a phenomenon only qualifies as a sign when it works as a sign, i.e., when it evokes a certain behavior (disposition and reaction) that can be scientifically studied and empirically validated. This definition is interesting for the field of architecture, because it opens up the possibility of studying the way in which the built environment (as sign vehicle) evokes certain behaviors in people. His best-known contribution, however, has to do with how he subdivides semiotics into three subfields: syntactics (dealing with the relations between signs), semantics (dealing with the relation between sign vehicle and object), and pragmatics (dealing with the relation between sign vehicle and interpretant).²⁶ His followers tended to either interpret this as three aspects of the same process of semiosis or as three different domains of semiotic research. Morris himself clearly stressed the interrelation between the three fields.

The intimate ties of the Anglo-American strand of semiotics with Pragmatism are indicative of its empirical bias and its attention to sign users. John L. Austin's (1911-1960) theory of "Speech Acts" can be seen as closely connected to these aspects of semiotics.²⁷ This theory posits that many language utterances, such as wishes or promises, are not only communicating a certain informative content but also have a prerogative effect on the relation between speaker and listener. Language indeed can also *act*, in that it changes the reality outside of language. These ideas obviously have relevance for nonverbal languages as well, such as for body language or body movements. That is what Edward T. Hall investigated in his 1966 publication *The Hidden Dimension*, where he developed his theory of "proxemics," explaining how people determine the distance they keep from one another (differentiating between, for example, intimate distances, personal distances, social distances, and public distances).²⁸ Hall's insights into the wide variety of meanings associated with distance in different cultural contexts have had a certain impact on architectural thinking, which we will deal with later in this chapter.

Scenes of architectural semiotics: Freewheeling on paradigmatic tracks

Umberto Eco's redemption of functionalism

Italian philosopher and semiotician Umberto Eco (1932-2016) was responsible for turning Continental semiology towards a reappraisal of Peirce, Morris, and others, widening the scope of semiotics with his 1968 book *La struttura assente*.²⁹ In this book, he also presented a theoretical frame for dealing with nonverbal languages, with a special focus on architecture. Eco situates the sign within a twofold

perspective of communication and signification, including codes and messages in the semiotic process. He steers away from endless epistemological polemics between Saussurean and Peircean semiotics by simply positioning all available bipolar and triadic definitions of the sign on a triangular scheme, with the signifier, the signified, and the referent (or equivalents such as sign vehicle, interpretant, object) occupying the three corners. Eco suggests that such a scheme defines a semiotic space that gathers all terminological disagreements in a coherent figure and simply situates his own semiotic concerns on the left side of that triangle, indicating his focus on the coded relation between signified and signified. He considers the debate regarding the “objective reality out there” as less relevant to a theory of signification and communication that focuses on meaning as a message to be transmitted, and not as an outside reality to be represented.³⁰ For him this is a matter of methodological choice, not epistemological truth. Moreover, this pragmatic maneuver gives him the opportunity to combine a Saussurean perspective with the much larger spectrum of phenomena, objects, and relations that Peirce handles, facilitating an interconnection between semiotics and architecture.

According to Eco, architectural signs consist of “sign vehicles” (signifiers) that have a pronounced, often complex materiality, and a range of meanings (signifieds). In light of certain codes (conventional systems linking specific signifiers to specific signifieds), architectural signifiers can “denote” precise functions as their meaning. The strictly functional meanings (Eco refers to “the primary functions”) of these signifiers can be extended, with successive meanings (“secondary functions”) obtained via “connotations” derived from other codes. Eco emphasizes that the primary, denotative functions are no less “symbolic” than the secondary ones and that the secondary connotative functions are no less “functional” than the primary ones.³¹ For example, one does not automatically know, on the basis of its form, how to use a simple staircase; rather, one has to learn this use. Hence the relation between form and function is not intrinsic but symbolic. The symbolism of a majestic staircase, on the other hand, functions as an indication of power and prestige and is thus able to modify the perception and use of that staircase. Considering function as meaning is central to Eco’s architectural semiotics. He quotes Barthes: –“every usage is converted into a sign of itself”³² – and refers to Sullivan’s “form follows function.”³³ This means that the architectural form should make the function possible and at the same time communicate that function, that is, make the function obvious, necessary, and attractive. In this way, Eco’s sign touches upon Austin’s notion of a “speech act”: in a certain way a form stimulates one to perform a function.

Eco’s approach makes room for the complex temporality of architecture by pointing out that the three dimensions of the sign – the material sign vehicle, the denotative first functions, and the connotative second functions – have each their own life spans, durations and rhythms of change.³⁴ Built forms may undergo

transformations; denotative primary functions may change, be disaffected, or reinstalled; connotative secondary functions may alter, become more prominent, or fade away. This results over time in a game of loss, recuperation, and substitution, and thus in a layering, densification, dilution, or shifting of forms, of denotative functions/meanings, and of connotative functions/meanings. In order to study a building's "history of signification," changes to its form, primary and secondary functions can be plotted along three separate timelines that together outline a complex semiotic narrative. The multiple time frames (immediacy, steady transformation, "longue durée") proper to the pace of construction, the histories of dwelling, and the life span of buildings bring in a kind of "timing of signification," an awareness of history, which offers an interesting amendment to the synchronic bias that often prevails in structural semiotics. At the same time, the separate timeline of form attributes a degree of autonomy to the architectural form. However, because of the semiotic association of sign vehicle and function, this autonomy is never an absolute one.

For Umberto Eco, signification is closely linked to communication, whereby architectural communication shows strong similarities to mass media, which work as "systems of rhetorical formulas" aimed at both convincing the mass of consumers and meeting their expectations.³⁵ Just like television or fashion, architecture is a part of everyday life. Hence people live with architecture without necessarily experiencing it on a conscious level. And yet, where mass media confirm and even reinforce existing social premises and prejudices, Eco is open for the possibility that architecture could succeed in distancing itself from social expectations and conventions in order to renew itself, to engage critically with social ideologies, and even to undermine them. In Eco's opinion, architecture functions according to the model of an "open language system" that relies upon both internal and external codes. He mentions technical and typological codes as examples of well-elaborated internal codes in architecture.³⁶ These codes can be denotative or connotative and can operate on both a syntactic and a semantic level. He grants these internal codes the capacity to innovate architecture in a technical, aesthetic, or functional sense. Achieving fundamental innovation, however, would only be possible by relying upon external codes that belong to the social or anthropological field outside architecture.

Eco refers to Hall's "proxemics" as an instance of such external cultural codes, which provide an underlying structure that conditions architecture.³⁷ This interpretation however evokes questions. Why does Eco refer to proxemics as an external code? Shouldn't one recognize that the logics of proxemics are intimately related to architectural settings and hence form a code internal to architecture? Hall refers after all to practices like "poking one's head in the door" in an office in Germany, or the table setting in a French restaurant. In his book *La struttura assente*, Eco criticizes many structuralists for mistakenly attaching an ontological reality to the

structures they detect in their analysis of cultural practices. He argues that it makes more sense to consider these structures as “absent” or to rather characterize them as methodological choices. If we follow him in this respect, shouldn’t we wonder whether he is making a similar mistake by considering proxemics as an underlying structure that ontologically exists outside of architecture? If this objection makes sense, it also makes sense to question Eco’s assumption that fundamental innovation in architecture always comes from sources outside of architecture.

It might be that Eco himself also evolved in his opinion regarding this question. At the end of his chapter on the semiotics of architecture, there is a passage on the design of Brasilia, which suggests that the “failure” of Brasilia was less the responsibility of its designers than of the briefs they received, which did not adequately predict societal evolutions: since the architects expertly designed sign vehicles that closely responded to these external briefs, they were not at fault. He wonders however whether it would not have been a better option for the architects to design “forms and dispositions flexible enough to provide for different meanings as warranted in the course of events.”³⁸ Here one can detect a suggestion that it might be possible for architecture to rely upon its own internal codes to develop, for example, innovative typologies that would be more open and that would allow for the accommodation of future societal contradictions, not yet taken into account in codes that are external to architecture.

Geoffrey Broadbent and the semiotics of design modes: Walking with Peirce

In the sixties and seventies architectural semiotics was not the only “nouvelle vague” in theory and criticism of architecture. Another wave, propelled by the hope of defining a design theory and methodology capable of dealing with the needs of society in full modernization, was articulated by, among others, Nigel Cross, Horst Rittel, Serge Chermayeff, Christopher Alexander, and Geoffrey Broadbent. Not surprisingly several scholars attempted to establish bridges between design theory and semiotics.

A case in point is British architect and educator Geoffrey Broadbent (b. 1955). In several texts Broadbent staked out four modes of design that he considered as fundamental.³⁹ In his view, “pragmatic design” is based on trial and error: the designer or builder experiments until a form is created that serves the intended purpose. Pragmatic design might be the oldest but still the most widespread design mode. Broadbent refers to the prehistoric shelter of mammoth hunters but also to today’s experimentation with new materials and construction modes. If one includes ad hoc construction and self-help housing all over the world, pragmatic

design is by far the predominant design mode. “Iconic design” relates to “fixed mental images,” consolidated after a long process of pragmatic design and considered, by social convention and common knowledge, apt to specific briefs. Broadbent refers to vernacular forms such as “the Eskimo’s igloo” but also to office buildings, “which become the contemporary fixed mental image for a generation of architects and clients.”⁴⁰ In “analogic design,” inspiration is sought in observable similarities with external images, forms, spatial arrangements, etc. Broadbent mentions Le Corbusier’s comment on the “visual analogy” between the roof of his Ronchamp Chapel and a crab shell. He also points to the “structural analogy” between the plan of three Frank Lloyd Wright houses, all three based on an identical organogram but each elaborated with a different geometrical pattern.⁴¹ Finally, “canonical or geometric design” uses an abstract system of relations based on geometric analogy with the human body, the cosmos, or mathematical figures.

Subsequently, Broadbent connects these four “fundamental” design modes with the three best-known sign types from Peirce’s classification.⁴² Peirce’s “icon,” in which the representamen (signifier/sign vehicle) refers to an object on the basis of potential visual or structural resemblance (“firstness”), is, according to Broadbent, especially relevant in the analogical and canonical design process. Peirce’s “index,” where that reference is based on a factual or causal link (“secondness”), would be active in pragmatic design, from ad hoc building to the experimental use of new materials and techniques. Peirce’s “symbol,” where the reference from the representamen towards the object is based on a rule or a convention, works in “iconic” designs based on “fixed mental images.” Broadbent observes that his “iconic design,” sustained by “fixed mental images,” has less to do with Peirce’s “iconicity” than with Peirce’s “symbol,” and changes the name from “iconic” into “typologic design.” Finally, Broadbent makes a rather debatable move by ranking his four modes of design in terms of the degree of creativity they require. Pragmatic design (working with indexes) figures on top, followed by analogic design (working with icons of visual resemblance), typologic design (based on fixed mental images), and finally canonic design (involving icons of structural resemblance).

Broadbent’s walk with Peirce makes an important move in the Anglo-American semiotics of architecture. The link Broadbent establishes between design methods and Peirce’s semiotics opens the possibility of an interesting theoretical interplay between these two fields. Unfortunately, Broadbent limits his evocation of Peirce to the index-icon-symbol triplet that articulates the relationship between representamen (sign vehicle/signifier) and object, ignoring the fact that this triplet only deals with one of the three relationships at work in Peirce’s triadic definition of the sign. Consequently, the impact of the “three modes of being” (firstness, secondness, and thirdness) at play in all three relationships that make the sign work as a sign, is, in Broadbent’s writings, limited to their appearance as index, icon, or

symbol. Equally absent are other concepts that could be of interest in the semiotics of design, such as “the unlimited semiosis” a sign is capable of generating and the “abductive” (“what if”) mode of reasoning that prevails in Peirce’s logic.

Nevertheless, Broadbent’s move remains highly significant. By linking different design modes to different classes of signs in which different “modes of being” are at work, Broadbent implies that different “modes of signification” operate in design. This, in turn, can be connected to the idea of “modes of production,” a concept that is mostly prevalent in (neo)Marxist theory and that refers to different systems of political economy (i.e., capitalism versus communism). Umberto Eco likewise uses the term *modes de production sémiotiques*, implying that the production of significations might follow some kind of logic that is reminiscent of the logic of political economy.⁴³ As we discuss later, this potential crossover between semiotics and neo-Marxism would become productive indeed in the hands of Jean Baudrillard.

Charles Jencks: “The Language of Postmodern Architecture.” Reloading the metaphor

In accordance with Eco, British architectural historian Charles Jencks (1939-2019) posited the semiotic ideas that architecture is a medium for mass communication and signification, that it functions as a “language” that uses different sign types, and that it can be encoded and decoded in different ways. Similar to Broadbent, Jencks recycled the Peircean sign triplet of Index, icon, and symbol, but in his interpretation this classification is not a categoric one. On the contrary, in accordance with Peirce he sees all signs as compound signs: they all have indexical, iconic, and symbolic characteristics.⁴⁴ However, following Jencks, in architecture, signs are far more indexical and iconic than in linguistics: in the postwar period, the symbolic (conventional) aspects of architectural signs have been impoverished and the iconic dimension has lost its deeper metaphorical levels. In Jencks’s opinion, modern architecture is therefore to be blamed “for its obsessive concentration on indexical meanings” that promotes “a banal and literalist life of simplified functionality.”⁴⁵ Modern architecture has thus squandered architecture’s ability to communicate in a meaningful and convincing way with the general public (a sentiment that echoes Choay’s earlier complaint about “semantic reduction”).

Jencks’s agenda clearly takes aim at the failures of modernism. He grants the term “postmodern” to architects who regard architecture as a visual language to be renewed and who consciously do this by encoding it “in a double way.” This “double-coding” allows on the one hand a small professional and cultural elite to enjoy “fine discriminations in a fast-changing language,” while on the other hand it affords the general public or local residents who are interested in “beauty, a

traditional ambience, and particular way of life” to identify themselves with the iconography of the postmodern building.⁴⁶

In Jencks’s discourse, the crucial importance of communication in all architecture, be it historical, modern, or postmodern, implies a “positive approach to metaphorical buildings.” His semiotic method consists of identifying and describing as precisely as possible the metaphoric load of a building. Jencks thus inventorizes and interprets the metaphors used and points to their layering and interaction. The nature and intensity of the metaphors identified vary from explicit to hidden, from literal to suggestive, from ridiculous to intriguing. A good example is Le Corbusier’s chapel in Ronchamp with its multiple references, subtle or caricatured — mother and child, ship, nun’s hat ... For Jencks this is a metaphorically highly suggestive building.⁴⁷ So too is the hotdog stall in the shape of a hotdog, which he sees as a building whose metaphors have been explicitly coded for the uploading of one intended meaning and for the efficient communication of that particular meaning.⁴⁸

Jencks most convincingly manages to present the power of metaphor in the semiotic analysis of some historic pieces of architecture, such as Gaudí’s Casa Batlló.⁴⁹ Here he succeeds in unraveling different layers of overt and covert meaning whereby the visual communication simultaneously deals with the exposure of commemorative emblems, the whispering of secret messages, and the framing of a forbidden ideology, all related to Catalonia’s nationalist struggle. But even here Jencks limits his reading to the most outward appearance of the building: the roof and the street façade. No mention is made of the astonishing typology of its decorated interior, its noble floor, its lightwell, its atrium, its loft, its roof landscape, etc. In Jencks’s semiotic understanding, the metaphor thus remains a somewhat superficial trope, closer to achieving a visual surprise and spectacular effect through the external shape and appearance of the building than to revealing less obvious and unnamed meanings that might reside in other architectural features.

No doubt Jencks’s emphasis on metaphoric predication in architecture serves his purpose of promoting postmodern architecture and dismissing an exhausted modernism. This can be seen as a proactive form of semiotics that works somewhat like operative history, in that it legitimizes and supports a specific trend in contemporary architecture.⁵⁰ In refocusing on the role of metaphor in architectural thinking and practice, Jencks ventured into a potentially highly productive realm of cross-fertilization between architecture and semiotics. Though Jencks’s own work allows only for a rather limited interpretation of the potentials of metaphorical semiosis, several authoritative voices from other fields engaging with semiotics suggest more intriguing perspectives on the possible role of metaphor in architecture.

In a fragment of *La Pensée Sauvage* (1962), anthropologist Claude Lévi-Strauss (1908–2009) narrates the story of Mr. Wemminck, a character from Charles Dickens’s *Great Expectations*.⁵¹ Mr. Wemminck is a dull cleric who meets his daily

work obligations with “official sentiment” and who at home takes care of his elderly father. He takes genuine pleasure, however, in the reconversion of his suburban cottage in Walworth into a miniature castle, with a drawbridge, a cannon that fires one shot every day at dinner time, and a kitchen garden surrounding the cottage that would help the residents to survive a siege. This self-built castle is the result, according to Lévi-Strauss, of a metaphorical operation that realizes a complex transfer between the syntagmatic chain of the (absent) castle and the one of the (present) cottage: transfer of objects (drawbridge, cannon), transfer of typological features (the surrounding kitchen garden as a moat-like protection apparatus), and transfer of etiquette rules (announcing dinner time). This gives rise to a creative tension that generates a new spatial form (a cottage-castle or a castle-cottage) as well as a new sentiment: the Walworth feeling.

The metaphor also plays a crucial role in the anthropology of Victor Turner (1920–1983), which studies rituals addressing fundamental changes (such as the transition of power, curing illness, crime and punishment, etc.) needed in a condition of conflict or crisis.⁵² Turner describes how in such rituals “multivocal symbols” (objects, creatures, costumes, rhythms, gestures ...) are brought together and evoke a chain of associations and related meanings. These components play a role in choreographies or role-plays that perform a “root metaphor” confronting the structural crisis with its antistructural counterpart, alternately confirming and negating the conflict, and thus generating a condition of suspension or liminality. This metaphorical play allows for a complex transfer of meaning in which the crisis of the traditional symbolic order is consecutively presented, dissolved, and reformulated, which allows to amend or to repair the damaged order.

In addition to anthropologists such as Lévi-Strauss and Turner philosopher and psychiatrist Jacques Lacan (1901-1981) offers interesting insights into metaphors. For Lacan metaphors are crucial in the interaction between the psychiatrist and the individual patient, because they somehow shed light on the shaky borderline between the conscious and the unconscious. Lacan develops a semiotic interpretation of this process on the basis of a reading of Saussure. He posits that the line that Saussure draws between signifier and signified is analogous to the one between “the conscious” and “the unconscious.” The conscious refers to the symbolic order with its syntagmatic or metonymic chain of signifiers, whereas the unconscious has to do with the domain of unreachable signifieds. The metaphor helps to bridge the line between both and to reach out to these hidden signifieds. The semantic gap in conscious reasoning might provoke a creative kindling of the metaphor, whereby a signified of the unconscious appears for a short while in conscious speech, as an unexpected signifier, creating a glimpse of new meaning.⁵³ This process helps the patient on his way to a *parole pleine* (full speaking), which is crucial for a healthy self-understanding of the subject.⁵⁴

Semiotics in its full breadth thus offers quite some interesting discussions of the metaphor and its role in creative signification, far beyond Jencks's initial opening. The metaphor apparently can help in redeeming semantic gaps, in dealing with social conflicts, and in the becoming of the subject. These hints however have not necessarily been picked up by architectural theory in the last decades of the 20th century. Still one can assume that buildings such as the Unité d'Habitation in Marseille, or the Centre Pompidou in Paris, or the Teatro del Mondo in Venice, or Parc de la Villette in Paris, or the Jewish Museum in Berlin might be more fully understood when their metaphorical operations are analyzed aided by such semiotical sources. Surely such analyses would clarify how their "creative kindling" managed to produce new spatialities and new affects, if not the promise of a *parole pleine* in architecture.

From Rossi to Derrida: Semiosis without semiotics. The city as text

There is no official common name to cover the work of the Italian neorealist/neorationalist *Tendenza*, spearheaded by Aldo Rossi (1931-1997), Giorgio Grassi (b. 1935), and Carlo Aymonino (1926–2010); the French *analyse urbaine* or "morpho-typology" associated with figures like Philippe Panerai (b. 1940) and Jean Castex (b. 1942); the Belgian "Reconstruction de la ville Européenne" proclaimed by a group of scholars and architects linked to the magazine AAM (*Archives de l'Architecture Moderne*); and the team in charge of the preparation of IBA 1984, directed by Josef Kleihues (1933-2004) and Hardt Hämer (1922-2012). They can all be gathered under the heading "the Architecture of the City approach," in honor of Aldo Rossi's seminal book launched in 1966 and repeatedly reprinted, reedited, and translated in the 1970s and 1980s.

Rossi's *Architecture of the City* accords the city as a whole, as well as its various architectural elements, a level of relative autonomy vis-à-vis other domains, such as culture or politics.⁵⁵ The city is seen as "architecture," consisting of built forms that shape its functional and social life, rather than the outcome of social factors. In that sense, the city honors a "function follows form" adagium, reversing the "form follows function" principle. The architecture of the city is thus considered as an important domain of material culture. Accordingly, the architectural form has its own formal logic that can be objectively analyzed, consistently theorized, and managed by design. The *longue durée* of the city's built history allows for the slow development of spatial relations and types while at the same time the city's architecture is continuously adapted and transformed. Central to this approach is the idea of type – a category of buildings sharing common characteristics. The architecture of the city is made of types of buildings and open spaces. Hence typology, the identification and classification of types and variants, becomes central to

analysis and design. This brings Rossi to posit the trope of “analogy,” defined as the relationship between types, as central to the architecture of the city. Rossi calls the historical European city “the Analogous City” – with analogy understood as both a mode of analysis and of design. He explains this notion by referring to an eighteenth-century “capriccio” painted by Canaletto, which presents a fictional urban scene in Venice composed as a collage of buildings, (paradigmatically) selected out of various contexts but all linked by analogy. Rossi calls this painting “a project.”

French urban analysis further invested in developing a consistent methodological frame for urban analysis. Revisiting various concepts of typology in history, typological analysis is brought together with morphology, the study of spatial relations and structures. The result is a coherent and workable method of “morpho-typology” that throughout the 1980s was taught in many schools in Europe. Urban analysis and urban history provided the typo-morphological principles that were also supposed to form the basis of design.⁵⁶

At first glance, this “architecture of the city” approach would seem rather distant from semiotics. Nevertheless, various lines of congruence between them can be identified. In the introduction of his book, Rossi underlines the “very evident analogies” between the study of the city and “linguistic studies,” mentioning how Saussure’s contributions to the development of linguistics could serve as a program for the development of urban science.⁵⁷ Indeed, the concept of type shares several characteristics with that of the Saussurean sign. Both Rossi and the French scholars refer to Quatremère de Quincy’s *Dictionnaire historique de l’Architecture* (1832) in order to define “type.” Quatremère’s concept of type is based on two complementary principles: on the one hand type stands for a set of heterogeneous, typological characteristics of an architectural object (form, style, plan, function, building components, construction mode, technology ...); on the other, it is articulated by a series of variants which together constitute the type.⁵⁸ This indeed corresponds, to a certain extent, to the Saussurean sign, because the type as a signifier gathers and associates typological characteristics as signifieds and furthermore acquires “value” in the paradigmatic differentiation of variants and in the syntagmatic combination with other types into an urban fabric (remember Saussure’s metaphor of the paper snippets). Secondly, typology and morphology can be compared to the two basic axes of language defined by Saussure, i.e., “selection” (of each sign from an associative or paradigmatic field of signs) and “combination” (of signs into a logic syntagma – remember the metaphor of the column).

The congruence between both fields highlights yet another aspect of the subject. Considering the morpho-typological method as an indirect, implicit form of architectural semiotics allows us to make a link with the shift from structuralism to post-structuralism. This shift took place under the influence of the work of, among others, Barthes, Lacan, Derrida, Althusser, Kristeva, Baudrillard, and Deleuze. In

semiotics' main discipline, language studies, the object of interest shifted from structural linguistics to psychoanalytical conversation, written language (*'écriture'*) and texts. The atomic aspect of the Saussurean sign, i.e., the association of a signifier and a signified, was no longer dominant. The signifier was liberated from its univalent association with a supposed signified and became the predominant element in the signification process (semiosis). This shift mobilized an understanding that presented an endless shifting of meaning, due to "a play of signifiers" that provoke a parade of signifieds, which in turn act as signifiers to highlight still other signifieds, in an endless chain of signification. In a post-structuralist understanding, signs and codes – but also analogies and metaphors – all work as different "modes of signification" or "modes of meaning production," eliciting temporary associations between signifiers and signifieds generating momentary meanings in different ways.

In language studies, complex or experimental writings by Dante, de Sade, Mallarmé, Balzac, Bataille, and others became privileged objects of semiotic research. Texts and text studies were seen as belonging to an intertextual field in which each text provided the potential material for new text production. For post-structural thinkers, reading – the production of a personal interpretation of a text – thus became a form of writing: reading/writing had to do with deciphering codes, tropes, and intertextual references, identifying the play of textual signifiers, pursuing the traces of these signifiers in their uncertain associations with (supposed) signifieds that disclose (provisional) meanings of the text. These signifieds turned out to be signifiers of yet other meanings, a process whereby ever new traces of meaning ensure that the finding of an ultimate meaning is endlessly postponed. Drawing on the verb *différer* ("to differ" but also "to postpone"), Jacques Derrida (1930-2004) called this process "*différance*" (instead of "*différence*").⁵⁹

The post-structuralist fascination with texts and intertextuality is akin to the methodology of urban analysis that approaches the city as a text to be deciphered, to be read, to be written. The "city as text" is indeed one of the root metaphors of the architecture of the city approach. Jean Castex invoked Henri Lefebvre in this regard: "He decided to approach the city objectively by studying the physical form of the city, reading it as the text from which the context, that is to say the social reality behind the city, could come to the fore."⁶⁰ Likewise Panerai and Castex called the urban analysis they made of Versailles a "reading of a city."⁶¹ Besides the metaphor of "the city as text," the metaphor of "the urban fabric" was called upon. This metaphor relies on the idea of the woven fabric, resulting from an interaction between warp and weft, generated by the double rhythm of the instant movement of the spool producing specific textures and colored motives interweaving the stable and robust warp. This "interweaving" is echoed in the interaction between the evervarying and diversifying types and the stable morphological frame outlining the public realm.⁶² Both the text and the fabric metaphor point to the necessity of multiple readings of

the city; as such, Panerai and Castex proposed a multitude of methods such as maps, measurements, graphical analysis, observations, conversations, texts, and walks in order to unlock the city's multiple codes and themes – that is, its modes of spatial articulation, inhabitation, and signification (i.e., the architecture of the city).⁶³

Rossi also reused the metaphor of the city as “collective memory,” caught in a never-ending dynamic of remembering and forgetting. As such, urban analysis became a form of “topo analysis,” digging up motives and events out of the urban unconscious – the forgotten or repressed repository of the city's life and history.⁶⁴ Urban analysis allows us not only to understand the physical form of the city but also to take up this physical form in order to interpret the interplay with the social and historical context. This context is, so to speak, stored in the built memory of the city, sedimented and engraved in its architecture.

Finally, in the “architecture of the city” approach, there is no caesura between urban analysis (as reading the city) and urban design (as writing the city), which is comparable to the reading-as-writing concept in post-structuralist semiotics. According to Rossi and Castex, analysis rather forms an integral part of the design process.⁶⁵ The concept of “the analogous city” hence informs a double program of urban analysis *and* urban design.

Semiotics and ideology: The political economy of sign and design

Poststructuralism not only celebrated the predominance of texts as the ultimate semiotic objects but also excelled in the critique of ideology as one of its most important concerns. This aspect allows us to uncover the relevance of the semiotic figure of knowledge in terms of social and societal critique. The nonmotivated character of the Saussurean sign with its arbitrary association and imaginary identification of signifier and signified had already opened a window for ideological practices and critique of ideology. The same applies to the notions of denotation and connotation. Both Barthes and Eco understood the critique of ideology as the unveiling of connotative meanings or secondary functions that are carried along with the initial denotation or the primary functionality of the sign.

The French neo-Marxist philosopher Louis Althusser (1918-1990) distinguished between, on the one hand, ideology as the imaginary representation of the real conditions of life (this is in line with the orthodox Marxist understanding of the term, which implies that the relations of production are presented to the powerless in a distorted way) and, on the other, ideology as the imaginary narrative that people adhere to when explaining their conditions of life (which is more in line with the neo-Marxist use of the term).⁶⁶ In the second definition, ideology is always present and plays an important role in the formation of a person's or a group's identity.

This process of identification operates in a similar way as the mirror stage (Lacan) in the life of a young child: children develop a sense of identity by identifying with roles, names, and behaviors offered by the social environment (parents, relatives, teachers).⁶⁷ Through the work of Althusser and others, psychoanalysis thus joined forces with neo-Marxist thought, and both become aligned with post-structuralism, as practiced, for example, by the authors connected to *Tel Quel* magazine.⁶⁸ These alliances allowed such authors to operate on both the level of highly sophisticated analyses of avant-garde texts by Joyce, Bataille, Artaud, and others *and* the level of social and societal critique.

One of the *Tel Quel* authors venturing in this field was philosopher Julia Kristeva (b. 1944), who has identified several modes of meaning production active in texts and calls them “idéologèmes.”⁶⁹ She searches in the “fenotext” (i.e., on the surface and performance level of the text) for traces of the idéologèmes that dominate meaning production on the level of the “genotext” (i.e., on the deep-structural and competence level of the text).⁷⁰ She sees the Saussurean sign as one of the most dominant idéologèmes, in that it is constituted as an arbitrary association of a signifier and a signified but also negates this arbitrariness by assuming an imaginary identification between signifier and signified.⁷¹ For Kristeva, the meaning production in the text is not a process without a subject. It is an “*expérience pratique*” (practical experience) through which the subject is able, in her writing/reading, to build critical awareness of the idéologèmes at work so as to increase *la parole pleine* and thus enhance the formation of her subjectivity.⁷²

A very interesting text that presents a crossover between critical theory, neo-Marxism, and semiotics in the field of architectural theory was produced by social scientist and philosopher Jean Baudrillard (1929-2007).⁷³ Baudrillard posited that in capitalist production, “things” (utensils, tools, whose exchange value traditionally corresponds to their use value) are turned into “products” (commodities) by subjecting them to a separation of their exchange value and use value. This is followed by a manipulation of both entities (dictated by industrial production and by price setting on the basis of profit making) and then a reunification on the basis of an imaginary identification between manipulated price and use value (referring to the assumption that more expensively priced items indeed offer more use value – which clearly is not always the case). Baudrillard extended this Marxist analysis by arguing that in late capitalist consumer society the generalization of consumption objects corresponds to a generalization of the semiotic logic of the Saussurean sign. This sign is also based on an arbitrary (not intrinsically motivated) association and imaginary identification (of signifier and signified). The consumption product likewise becomes a sign-object with its own “exchange value/sign” as signifier and “use value/sign” as signified.

This mechanism underpinned what Baudrillard now labeled as “the political economy of the sign.” It received an important theoretical and cultural impetus

from “the Bauhaus” (Baudrillard clearly used the term as a *pars pro toto* for modernism). In Baudrillard’s view, all possible values of an object, all its possible fulfillments of needs, requests, and desires, are reduced by the Bauhaus theory to two rationalized values: aesthetics and utility. Design artificially isolates these two values from each other in order to manipulate them and to reunite them after manipulation into one unsuspected and uncontested entity that is presented as rational, useful, natural, and true.⁷⁴ The Bauhaus in fact codified and promoted the denotative function of an object as that which is pure, beautiful, true, and desired, whereas any form of connotation was blamed and rejected as redundancy, uselessness, ugliness, falsification, kitsch, and fashion. In Baudrillard’s view, the Bauhaus thus aimed at a synthesis between form and function, between the beautiful and the useful, between art and technology, between the superstructure of form and meaning and the social and technical infrastructure achieved by the industrial revolution, presenting itself “as a second revolution achieving the industrial revolution and solving the contradictions left behind by the latter.”⁷⁵

However, in Baudrillard’s view, the Bauhaus saga does not end at this point. Two semiotic miscalculations break the power of the Bauhaus impetus. First, the distinction between denotation as “true” and connotation as “false” was an ideological construction, “a metaphysical fable,”⁷⁶ a “superior myth.”⁷⁷ That fable or myth was not solid enough to continue to rule the game of consumption. Behind “the metaphysical fable” proliferated a general fetishism that marked continuous new domains of consumption: money, beauty care, pets, mass media, body culture, sex and nakedness, the celebration of holiday and sun, the cultivation of gender difference, the glorification of the accomplished subject, the appropriation of a domesticated unconscious, etc. The second semiotic miscalculation reinforced the first one: “The Bauhaus and design claimed to control the process by mastery of the signifieds (use value based on the ‘objective’ evaluation of functions), but in fact it is the play of signifiers (a play dictated by exchange value) that took the lead. But that play is unlimited and escapes all control.”⁷⁸ As a consequence of both semiotic miscalculations, the Bauhaus functionalism, “the superior myth,” collapsed for the sake of fashion, which does not care about denotative truth or beauty or correctness but is fully engaged in playing the game of connotation.⁷⁹ Having lost any substantive or intrinsic objective, fashion plays a game of change for the sake of change – a play of pure signifiers. The age of the signified and of “pure” functionalism in design and architecture is over.⁸⁰

With this provocative analysis, Baudrillard attempted to disrupt the ideological veil that, in his view, was intrinsic to the Bauhaus (read modernism). Like Manfredo Tafuri, in his contemporary essay on *Architecture and Utopia*⁸¹ Baudrillard pinpointed the structural and ideological analogies between modernist theories on the one hand and the functioning of capitalism on the other, in order to enhance

architects' awareness of their embeddedness within socioeconomic structures that they were unlikely to escape. Neither Baudrillard nor Tafuri saw at that point any opportunities for architects to develop an architectural practice that might critically engage with this societal embeddedness and thus dismissed the relative autonomy that Rossi had granted to the architecture of the city and that Eco evoked with his call for an "open language system" capable of dealing with future social contradictions.⁸² This critical engagement was nevertheless something that many architects in the 1970s aimed for; for some of them their interest in semiotics was generated precisely by that ambition.

From Chomsky, over Derrida to Deleuze: Peter Eisenman's restless quest

Around 1960 a new semiotic concept overwhelmed Anglo-American linguistics and semiotics. MIT professor and linguist Noam Chomsky (b. 1928) provoked, with his *Syntactic Structures* (1957) and his *Aspects of the Theory of Syntax* (1965), a paradigmatic revolution in American language studies, comparable to the structuralist awakening caused by Saussure and Levi-Strauss on the European continent.⁸³ Chomsky's hypothesis of inborn "deep structures" that provide each of us with a "competence" of language acquisition, and that via "transformational rules" generate "surface structures" with a "performance of generating an unlimited series of grammatically correct sentences," echoed far beyond the realm of linguistics. Chomsky's attempts to discover that so-called Transformational Generative Grammar inspired both great enthusiasm and deep skepticism but also activated animated debates among linguists and philosophers.

Chomsky's ideas greatly inspired architectural theory and semiotics and obtained a particular significance for New York-based architect Peter Eisenman (b. 1928).⁸⁴ Eisenman thought that a universal grammar of architecture would be a significant step in the necessary project of regenerating the moribund modernism. He therefore embarked on a series of designs for houses (labeled House I, House II, etc.) in which he searched for abstract elements of form and for syntactic operations that would be universally applicable. Such a design process would exclude all semantic interpretations in terms of architectural history and modes of inhabitation and would not generate any meaning at all. The design process unfolds, within a perfect Euclidian cubic void, an astonishing series of geometric transformations of lines, planes, and volumes, from (supposed) deep-structural to surface level.⁸⁵ Although Eisenman explicitly intended to avoid semantics, the emergence of meaning proved to be inevitable: indeed the very operations that Eisenman performed on the basis of a Chomskyan logic became famous and thus turned into cultural

(hence meaningful) practices in themselves. Moreover, some of the houses were inhabited, which also implies that the interaction between house and residents generated processes of signification.⁸⁶ One could therefore hypothesize that Eisenman practices a splendid form of iconic design (in Broadbent's terms) by metaphorically transposing the linguistic idea of "universal grammar" to architecture and by morphing it into an architectural image in order to fill the semantic gap left by worn out modernism.

The Chomskyan wave however did not last long, as post-structuralism made its way across the Atlantic under the form of "French Theory."⁸⁷ In particular, Derrida's

"deconstruction," closely followed by Deleuze's "fold," hit the imagination of theorists and designers, including Peter Eisenman. One is left to wonder, however, how much effort was spent by architects on thoroughly understanding concepts such as "deconstruction" or "folding," and on thoughtfully interpreting these in terms of architecture. A lot of what came to be known as "deconstructivist" architecture, for example, seems to boil down to an almost literal rendering of the idea of deconstruction in an idiom of crooked corners, crumpled walls, and uneven floors that generate illusory suggestions of instability and vicissitude.⁸⁸ A similar straightforward "morphing" of epistemology into architecture led to an architecture of undulating walls and floors, pleated and continuous surfaces that supposedly evoked "the fold" theorized by Deleuze.⁸⁹ That the Derridean or Deleuzian inspiration sometimes resulted in remarkable and thus meaningful architecture such as Tschumi's Parc de la Villette, Libeskind's Jewish Museum, or Eisenman's Alteka Office Building has more to do with the design skills of these architects than with the semiotic soundness of the philosophy-aesthetic morphing that generated their architectural appearance.

Somewhere halfway in his astonishing series of paradigmatic evocations, Eisenman declared an absolute need for the radical rethinking of architecture itself and of its semiotics of representation.⁹⁰ His declaration echoed the post-structuralist semiotics of Baudrillard and Kristeva in putting forth the criticism that twentieth-century architecture was dominated by a classical conception of the sign, based on the "fictions" of representation, reason, and history. Eisenman rather advocated a "non-classical architecture," which would unmask itself as fiction by playing a game of "dissimulation" instead of "simulation." While "simulation" aims to eliminate the difference between reality and illusion so that the illusion can present itself as reality and hence replace it, "dissimulation" would leave the difference between reality and illusion unaffected (just like a mask does, because it does not deny that the wearer of the mask is not the being or the thing that the mask evokes). For Eisenman, this non-classical architecture is like a mask in that it "is no longer a certification of experience or a simulation of history, reason, or

reality in the present. Instead, it may more appropriately be described as an *other* manifestation, an architecture *as is*, now as a fiction. It is a representation of itself, of its own values and internal experience.”⁹¹ If one includes Eisenman’s own iconic representations of Chomsky, Derrida, or Deleuze in the history of representation, his move towards “dissimulation” might be seen as an interesting auto-critique. Unfortunately, however, it seems that non-semiotic dissimulation is not easily articulated into architecture. Maybe it is simply not possible to escape semiotics.

Eisenman’s remarkable quest of paradigms illustrates both the paradigmatic volatility of late twentieth-century architectural theory and its eagerness to participate in the articulation of a relevant theoretical frame capable of dealing with the increasingly complex challenges of culture and society. It thus offers an interesting figure of knowledge, which operates as an architectural manifestation of different strands of semiotics, sometimes trying to outdo semioticians in a quest for radical moves⁹² but ultimately bound by the social, economic, psychological, and technological limitations that delimit architecture’s playing field. Nevertheless the simple suggestion that architecture *can* perform radical and critical gestures by concentrating on its own formal logic (i.e., its relative autonomy) represents a highly significant moment in late twentieth-century theory and practice – even when later critics wondered whether these consecutive experiments with different games of signifiers and modes of paradigmatic morphing indeed succeeded in performing the societal critique they were supposedly aiming at.⁹³

Critical pragmatism, resetting semiotics

The two paradigmatic lines outlined above offer a certain coherence in the rhizomatic story of architectural semiotics from 1960 to 1980. In the Saussurean line, the arbitrary association and imaginary identification of the signifier and signified – together with the roles of denotation and connotation in the process of signification – made semiotics suitable for ideology critique and hence offered an anchorpoint for a sophisticated criticism that addressed architecture as a formal language *and* as a carrier of ideological functions (Tafuri, Baudrillard). The Saussurean line was also characterized by a development from structuralism to post-structuralism, from the semiotics of signs and syntax to the semiotics of signifiers and texts, from the identification of meaning to the tracing of multiple and provisional meanings, thus moving closer to urban analysis and urban design, a movement facilitated by the “city as text” metaphor, by the reading-writing / analysis-design nexus, and by the creative role of metaphor in language and design (Rossi, Panerai, Castex).

The Anglo-American line on the other hand started with the pragmatist semiotics of Peirce that opened up to a much wider field of signification. All objects

and phenomena, all verbal and nonverbal languages, all elements of material and immaterial culture can figure in Peirce's complex definition of the sign and its process of semiosis. The purpose of signification is provoking an effect on the basis of "what if" hypotheses. Therefore each Peircean sign sets in motion an appropriate combination of three "modes of being" (potentiality, factuality, conventionality). Within this pragmatist set up, scholars in architecture such as Jencks and Broadbent made attempts to link "modes of design" to the Peircean modes of being presented by the sign classes: icon, index, and symbol. Starting from this Peircean frame, Jencks also unlocked the significance of metaphors in architecture, without however fully realizing the analytical and critical potentials that this approach might allow for.

A major conceptual move of interconnection between the Anglo-American and Saussurian lines was effectuated by Eco, who defined function as meaning. Eco attributed a relative autonomy to the architectural form defined as a complex sign vehicle that communicates and promotes denotative and connotative functions to which it is associated by cultural codes. For him, semiotics of communication involves performance of function and as such comes close to the pragmatist semiosis aiming at effect. We found another instance of interconnection between both lines in the work of Eisenman, who at first was deeply influenced by Chomsky's generative grammar and later became fascinated by post-structuralist authors such as Derrida.

Revisiting the brief but complex history of architectural semiotics brought us to a conclusion that is quite different from the one often reached by other observers. It seems to us that the "forest of symbols" that we explored was far too easily discredited as a kind of wasteland by authors eager to introduce fresh concepts or inspirations.⁹⁴ If the quick sequence of fascination, disappointment, dismissal, and replacement was particularly striking with respect to this specific episode of architectural theory, the phenomenon is not limited to semiotics. It rather reveals a *modus operandi* that persists within the discipline of architecture – that of borrowing paradigms from other fields (linguistics, cybernetics, philosophy, sociology, anthropology ...) which are rapidly applied, consumed, and thrown away. The appropriation of each new paradigm is often limited to its most accessible text passages, to the most eloquent quotes, or to a collection of suggestive terms. Little intellectual energy seems to be spent on a careful translation of borrowed key concepts, which might lead to a truly creative metaphorical transfer rather than a fashionable gimmick.

Architectural semiotics is one of the paradigms that fell victim to the "postcriticality" wave that hit architecture around the turn of the century.⁹⁵ Postcriticality blamed postwar paradigms for charging architecture with an overdose of critical, epistemological, ideological, and social concerns that deviated the discipline from

its core business of projecting and building. Semiotics was a prime target in the line of fire because of its emphasis on representation, its focus on deciphering and interpreting, its search for the ever-elusive meaning, its obsession with difference and contradiction, its hermetic jargon, and its lack of tangible results. In its war on theory and criticism the postcriticality paradigm advanced some interesting positions, including both a justified attempt to restore the balance between autonomy and heteronomy in architecture and a renewed focus on the importance of design and projects. On the other hand, postcritical positions often showed a reductive interpretation of the contributions of theory and criticism (including semiotics), and the postcritical “projective turn” often implied a problematic denial of the social and political implications of projects.

It is our contention therefore that rather than dismissing successive paradigms out of hand – including both semiotics and postcriticality – architectural theory would do well to recognize, evaluate, and valorize its own recent history of thought and practice. In our opinion, our retrospective of semiotic paradigms and scenes did not unveil a collection of intellectual failures but rather a sequence of promising concepts, interrupted reasonings, unaccomplished adaptations, and semi-results. All this provides materials for critical evaluation, selective recycling, and further processing in the light of the actual condition of the discipline and its present challenges. For us, the two lines that we explored continue to offer valuable insights and the investigation of their potential interconnection remains a worthwhile intellectual challenge. Whereas we appreciate the ideology critique and political savviness that came out of the Continental line, we also value the common-sense logic and the potential social sensibility of Anglo-Saxon Pragmatism. For Umberto Eco, the different paradigmatic tracks do not really contradict each other but rather outline a common space of semiotic thinking and practice. Within such space reloading semiotics could be a meaningful program. “What if” crossovers and creative interplays between (post)structuralism and pragmatism, between the linguistic turn and the projective turn, between critical thinking and projective matters of concern,⁹⁶ could give form and content to something yet to be named (“the semiotics of critical projectivism”?, “design for critical pragmatism”?). A similar attitude seems to be shared by Joan Ockman, in her quest for theoretical support for contemporary architecture in American philosophical Pragmatism.⁹⁷ We think Ockman’s seminal effort merits continuation. We indeed remain convinced that architectural practice, architectural history, and architectural education continue to be in need of the critical reflection provided by an architectural theory that recognizes, appreciates, and build upon its own intellectual legacies. The efforts to embrace a critical Pragmatism and to reset semiotics are dearly needed for an architectural practice that acknowledges its responsibilities in moving towards an ecologically sound and non-oppressive environment.

Notes

1. Following Umberto Eco's suggestion, we use the term "semiotics" as a general category that covers all contributions to "sémiologie" and semiotics. Occasionally the term "sémiologie" is used in its historical particularity. See: Umberto Eco, *La structure absente: Introduction à la recherche sémiotique* [1968] (Paris: Mercure de France, 1972), 11.
2. Françoise Choay, "Urbanism & Semiology," in *Meaning in Architecture*, eds. Charles Jencks and George Baird (London: Barrie & Rockliff, The Cresset Press, 1969), 27-39, 34.
3. Alexander Tzonis and Liane Lefaivre, *Classical Architecture. The Poetics of Order* (Cambridge, MA: MIT Press, 1986).
4. Mart Stam, "Das Mass, das richtige Mass, das Minimummass," [1929] in *Neues Bauen, neues Gestalten: Das Neue Frankfurt/die neue Stadt. Eine Zeitschrift zwischen 1926 und 1933*, ed. Heinz Hirdina (Berlin: Elefant Press, 1984) 215-16, 216.
5. Adolf Loos, "Potemkin City" [1898], *Spoken into the Void: Collected Essays, 1897-1900* (Cambridge, MA: MIT Press, 1982), 95-96; Adolf Loos, "Ornament and Crime" [1908], in *Ornament and Crime: Selected Essays* (Riverside, CA: Ariadne Press, 1998) 167-76.
6. Louis Sullivan, "The Tall Office Building Artistically Considered," [1896], in *The Public Papers*, ed. Robert Twombly (Chicago: University of Chicago Press, 1988), 104-13; Loos, "Ornament and Crime"; Amédée Ozenfant and Charles-Edouard Jeanneret, "After Cubism," [1918], in *L'Esprit nouveau: Purism in Paris, 1918-1925*, ed. Carol S. Eliel (New York: Harry N. Abrams, 2001).
7. John Summerson, "The Case for a Theory of Modern Architecture," *RIBA Journal* (June 1957): 307-10, 310.
8. The forest metaphor is borrowed from: Victor Turner, *The Forest of Symbols: Aspects of Ndembu Ritual* (Ithaca: Cornell University Press, 1967).
9. Louis Martin, "The Search for a Theory in Architecture: Anglo-American Debates 1957-1976," (PhD diss., Princeton University, 2002); Louis Martin, "Semiotics and Architecture Revisited," in *The Architect and the Public: On George Baird's Contribution to Architecture*, ed. Roberto Damiani (Rome: Quodlibet, forthcoming).
10. Ferdinand de Saussure, *Cours de linguistique générale* [1916] (Paris: Payot, 2016), 85-88, 151-56.
11. Roland Barthes, "Éléments de sémiologie," *Communications* 4, no.1 (1964): 91-135; Saussure, *Cours*, 215-18.
12. Barthes, "Éléments de sémiologie." Barthes refers to Saussure's notions of "syntagm" and "association," redefined as "syntagm" and "paradigm" by the Russian linguist to the tropes of "metonymy" and "metaphor." Roman Jakobson, "Deux aspects du langage et deux types d'aphasie," in *Temps Modernes*, no. 188 (Jan 1962): 853 ff.

13. Saussure, *Cours*, 230-32; see also Umberto Eco, "A Componential Analysis of the Architectural Sign /Column/," in *Signs, Symbols and Architecture*, eds. Geoffrey Broadbent, Richard Bunt, and Charles Jencks (New York: Wiley, 1980), 213-32.
14. Roland Barthes, *Mythologies* [1957], trans. Annette Lavers (London: Paladin, 1972). Barthes refers, for the concepts of denotation and connotation, to Louis Hjelmslev, *Prolegomena to a Theory of Language* [1943] (Baltimore: Indiana University, 1953).
15. Roland Barthes, "Sémiologie et Urbanisme," *L'Architecture d'Aujourd'hui*, no. 153 (1971): 11-13.
16. Charles Sanders Peirce, *The Collected Papers of Charles Sanders Peirce*, eds. Charles Hartshorne, Paul Weiss, and Arthur W. Burks (Cambridge, MA: Harvard University Press, 1931-1935, 1958), 2.227-228; 2.243; 5.475-476; 5.483; 5.488; 8.332; see also Eco, *La structure absente*, 22-23.
17. Peirce, *Collected Papers*, 1.23-26; 1.418-420; 8.328-331.
18. *Ibid.*, 2.243-252.
19. Catherine Legg and Christopher Hookway, "Pragmatism," in *The Stanford Encyclopedia of Philosophy*, ed. Edward N. Zalta (Stanford University: Metaphysics Research Lab, Spring 2019), <https://plato.stanford.edu/archives/spr2019/entries/pragmatism/>.
20. This effect can be "emotional," "energetic," or "logical." Peirce, *Collected Papers*, 5.475-76; 8.338-39.
21. Jacques Derrida, *Positions* (Paris: Minuit, 1972), 35-41.
22. The pragmatic maxim states: "Consider what effects that might conceivably have practical bearings, we conceive the object of our conception to have. Then, our conception of these effects is the whole of our conception of the object." Peirce, *Collected Papers*, 5.402.
23. Peirce, *Collected Papers*, 4.536; 8.315. See also Umberto Eco, *Le Segno* [1972] (Brussels: Labor, 1988), 251-53; and Wilhelmus A. De Pater and Pierre Swiggers, *Taal en teken: Een historisch-systeematische inleiding in de taalfilosofie* (Leuven: Leuven University Press, 2000), 125, 130.
24. Peirce, *Collected Papers*, 2.619-25; 2.632-44.
25. De Pater and Swiggers, *Taal en teken*, 137-51; Charles W. Morris, *Writings on the General Theory of Signs* (The Hague: Mouton, 1971), 17, 173-76, 224-27, 301-3, 340, 366, 401-5.
26. Morris, *Writings*, 21-22, 301-2.
27. John Langshaw Austin, *How to Do Things with Words: The William James Lectures Delivered in Harvard University in 1955*, eds. James Opie Urmson and Marina Sbisa (Cambridge, MA: Harvard University Press, 1975).
28. Edward T. Hall, *The Hidden Dimension* (Garden City: Doubleday, 1966).
29. Peter Bondanella, *Umberto Eco and the Open Text: Semiotics, Fiction, Popular Culture* (Cambridge, UK: Cambridge University Press, 1997). The history of Eco's relevant publications is rather complex. He first published *La Struttura Assente: Introduzione alla ricerca*

- semiologica* in Italian (Milan: Bompiani, 1968). Four years later this book appeared in a French translation as *La structure absente*. Section C of this book (“La fonction et le signe: Sémiotique de l’architecture”) was translated in English as Umberto Eco, “Function and Sign: The Semiotics of Architecture,” in *Signs, Symbols and Architecture*, 11-69. Another relevant book is Umberto Eco, *Segno* [1972] (Milan: Arnoldo Mondatore, 1980), translated in French as *Le Signe*.
30. Eco, *Le Signe*, 35-41.
 31. Eco, “Function and Sign”, 15-27.
 32. Ibid., 13.
 33. Sullivan, “The Tall Office Building”.
 34. Eco, “Function and Sign”, 25-34.
 35. Ibid., 41-42.
 36. Ibid., 38-50.
 37. Ibid., 50-57.
 38. Ibid., 58-61.
 39. Geoffrey Broadbent, “A Plain Man’s Guide to the Theory of Signs in Architecture,” *Architectural Design* no. 7-8, 1977; Geoffrey Broadbent, “The Deep Structures of Architecture” [1974], in *Signs, Symbols and Architecture*, 119-68; Geoffrey Broadbent, “Building Design as an Iconic Sign System” [1974], in *Signs, Symbols, and Architecture*, 311-31.
 40. Broadbent, “The Deep Structures,” 139.
 41. Broadbent, “Building Design,” 315.
 42. Ibid., 314-30.
 43. Eco, *Le Signe*, 165, 172-83.
 44. Charles Jencks, “The Architectural Sign,” in *Signs, Symbols and Architecture*, 71-118, especially 102-3.
 45. Jencks, “The Architectural Sign,” 107.
 46. Charles Jencks, *The Language of Post-Modern Architecture* (London: Academy Editions, 1977), 8.
 47. Jencks, *The Language*, 48.
 48. Ibid., 40-52.
 49. Jencks, “The Architectural Sign”, 92-95.
 50. Manfredo Tafuri, *Theories and History of Architecture* (London: Granada, 1980).
 51. Claude Lévi-Strauss, *La Pensée Sauvage* (Paris: Plon, 1962), 26, 198-99.
 52. Viktor Turner, *Dramas, Fields and Metaphors. Symbolic Action in Human Society* (London: Cornell University Press, 1974), 13-17, 25, 31, 33-52, 202, 248, 272-99.
 53. Jacques Lacan, “L’instance de la lettre dans l’inconscient ou la raison depuis Freud,” in Jacques Lacan, *Ecrits* (Paris: Seuil, 1966), 493-518.
 54. Lacan, “Fonction et champs de la parole et du langage,” in *Ecrits*, 247-59.

55. Aldo Rossi, *L'architettura della città* (Padua: Marsilio, 1966); English trans. *The Architecture of the City* (Cambridge, MA: MIT Press, 1982); French trans. *L'architecture de la ville* (Paris: L'Esquerre, 1981).
56. Giorgio Grassi, "Architekturprobleme und Realismus," *Archithese* (19: Realismus in der Architektur), 1976, 18-24; Léon Krier, "The Blind Spot," *Architectural Design* no. 4, 1978, 219-21; Jean Castex, "Enjeu et nécessité," in *Eléments d'analyse urbaine*, ed. Philippe Panerai et al. (Brussels: AAM, 1980), 710; Philippe Panerai, "Typologies," in *Eléments d'analyse urbaine*, 73-108; Christian Devillers, "Typologie de l'habitat & morphologie urbaine," *L'Architecture d'Aujourd'hui* no. 174, July/August 1974, 18-22.
57. Rossi, *L'architecture de la ville*, 10.
58. Ibid., 25-27; Panerai, "Typologies."
59. Derrida, *Positions*, 35-41.
60. Castex, "Enjeu et nécessité," 7. Castex is referring to Henri Lefebvre, *Le droit à la ville* (Paris: Anthropos, 1968). A few years after Lefebvre, Greimas formulated a similar text-city metaphor. See A. J. Greimas, *Sémiotique et sciences sociales* (Paris: Seuil, 1976), 140-41.
61. Jean Castex, Patrick Céleste, and Philippe Panerai, *Lecture d'une ville: Versailles* (Paris: Moniteur, 1980).
62. The metaphor of the "urban fabric" is a common one. For the specific articulation into "warp" and "weft," see SAR-73, *Het methodisch formuleren van afspraken bij het ontwerpen van weefsels* (Eindhoven: Stichting Architecten Research, 1973); Castex et al., *Lecture d'une ville*, 9-10; Alain Borie, Pierre Micheloni, and Pierre Pinon, *Forme et déformation des objets architecturaux et urbains* (Paris: Centre d'études et de recherches architecturales, 1978), 27.
63. Panerai et al., *Eléments d'analyse urbaine*.
64. Rossi, *Architecture de la ville*, 171-73 (Rossi refers to Maurice Halbwachs, *La mémoire collective* [Paris: PUF, 1950]); Greimas, *Sémiotique*, 129-31 (Greimas discusses "une sémiotique topologique").
65. Rossi, *Architecture of the City*, 219-20; Castex, "Enjeu et nécessité," 8-9.
66. Louis Althusser, "Idéologie et appareils idéologiques d'état," in Louis Althusser, *Positions* (Paris: Editions Sociales, 1976), 67-125.
67. Jacques Lacan, "Le Stade du miroir comme formateur de la fonction du Je," in *Ecrits*, 93-100.
68. Julia Kristeva, "La sémiologie: Science critique et/ou critique de la science," in Tel Quel, *Théorie d'ensemble* (Paris: Seuil, 1968), 80-93.
69. Julia Kristeva, *Sémiotikè: Recherches pour une sémanalyse* (Paris: Seuil, 1969), 60.
70. Julia Kristeva, "Sémanalyse et production de sens," in A. J. Greimas et al., *Essais de sémiotique poétique* (Paris: Larousse, 1972), 207-34.
71. Julia Kristeva, *Sémiotikè*, 83-89.
72. Julia Kristeva, *La révolution du langage poétique* (Paris: Seuil, 1974), 176-85.

73. Jean Baudrillard, *Pour une critique de l'économie politique du signe* (Paris: Gallimard, 1972), 172-99; 229-48; an English version was published as "Design and Environment: Or, the Inflationary Curve of Political Economy," in *The Universitas Project*, ed. Emilio Ambasz (New York: Museum of Modern Art, 2006), 50-65; see also Matthew Holt, "Baudrillard and the Bauhaus: The Political Economy of Design," *Design Issues* 32, no. 3 (Summer 2016): 55-66, https://doi.org/10.1162/DESI_a_00399.
74. Baudrillard, *Pour une critique*, 235, 245.
75. Ibid., 231.
76. Ibid., 245.
77. Ibid., 196. The full citation reads: "Loin d'être le terme objectif auquel s'oppose la connotation comme terme idéologique, la denotation est donc, puisqu'elle naturalise ce process meme de l'idéologie, le terme le plus idéologique...le mythe supérieur dont parle Barthes." The reference is to Roland Barthes, *S/Z* (Paris: Seuil, 1970), 16.
78. Ibid., 247.
79. Ibid., 246.
80. Ibid., 248; Jean Baudrillard, "Modernité," in *La modernité ou l'esprit du temps* (Paris: L'Esquerre, 1982), 28-31.
81. Manfredo Tafuri, *Architecture and Utopia: Design and Capitalist Development* [1973] (Cambridge, MA: MIT press, 1990).
82. Manfredo Tafuri, "Toward a Critique of Architectural Ideology" [1969], in *Architecture Theory Since 1968*, ed. Michael Hays (Cambridge, MA: MIT Press, 1998), 6-35 (this essay was further developed into the 1973 book *Architecture and Utopia*); Eco, "Function and Sign," 58-61.
83. Noam Chomsky, *Syntactic Structures* (The Hague: Mouton, 1957); Noam Chomsky, *Aspects of the Theory of Syntax* (Cambridge, MA: MIT Press, 1965).
84. Broadbent, "The Deep Structures"; Mario Gandelsonas and David Morton, "On Reading Architecture," in *Signs, Symbols and Architecture*, 243-74; Peter Eisenman, *Houses of Cards* (Oxford: Oxford University Press, 1987); Peter Eisenman, *Eisenman Inside Out. Selected Writings 1963-1988* (New Haven: Yale University Press, 2004).
85. Amir Djalali, "Eisenman beyond Eisenman: Language and Architecture Revisited," *The Journal of Architecture* 22, no. 8 (2017): 1287-98, <https://doi.org/10.1080/13602365.2017.1394350>.
86. Suzanne Frank, *Peter Eisenman's House VI: The Client's Response* (New York: Watson-Guptill, 1994).
87. François Cusset, *French Theory: Foucault, Derrida, Deleuze et cie et les mutations de la vie intellectuelle aux Etats-Unis* (Paris: La Découverte, 2003); François Cusset, *French Theory: How Foucault, Derrida, Deleuze, & Co Transformed the Intellectual Life of the United States* (Minneapolis: University of Minneapolis press, 2008).
88. Philip Johnson and Mark Wigley, *Deconstructivist Architecture* (New York: Museum of Modern Art, 1988).

89. Douglas Spencer, "The New Phantasmagoria: Transcoding the Violence of Financial Capitalism," in ed. Nadir Lahiji, *The Missed Encounter of Radical Philosophy with Architecture* (London: Bloomsbury, 2014), 79-93.
90. Peter Eisenman, "The End of the Classical: The End of the Beginning, the End of the End," *Perspecta*, 21 (1984): 154-72.
91. *Ibid.*, 167.
92. Céline Bodart analyses, in her PhD dissertation, the collaboration – or lack thereof – between Peter Eisenman and Jacques Derrida in their attempt to come up with a design for one garden in Parc de la Villette. See Céline Bodart, "Architecture et Déconstruction: Remises en jeu d'une rencontre: raconter, traduire, hériter" (doctoral dissertation presented at the Université 8 de Paris and the Université de Liège, 2018).
93. Mary McLeod, "'Other' Spaces and 'Others,'" in *The Sex of Architecture*, ed. Diana Agrest, Patricia Conway, and Leslie Kanes Weisman (New York: Harry N. Abrams, 1996), 15-28; Robert Somol and Sarah Whiting, "Notes around the Doppler Effect and Other Moods of Modernism," *Perspecta* 33 (2002): 72-77, <https://doi.org/10.2307/1567298>.
94. Deborah Hauptmann and Andrej Radman, "Asignifying Semiotics as Proto-Theory of Singularity: Drawing Is Not Writing and Architecture Does Not Speak," *Footprint* 8, no. 1 (2014): 1-12, <https://doi.org/10.7480/footprint.8.1.794>; Mark Foster Gage, "Deus Ex Machina: From Semiology to the Elegance of Aesthetics," *Architectural Design* 77, no. 1 (2007): 82-85, <https://doi.org/10.1002/ad.401>.
95. Ole W. Fischer, "Architecture, Capitalism, and Criticality," in *The SAGE Handbook of Architectural Theory*, eds. C. Greig Crysler, Stephen Cairns, and Hilde Heynen (London: SAGE, 2012), 56-70.
96. Bruno Latour, "Why Has Critique Run out of Steam? From Matters of Fact to Matters of Concern," *Critical Inquiry* 30, no. 2 (2004): 225-48, <https://doi.org/10.1086/421123>.
97. Joan Ockman, ed., *The Pragmatist Imagination. Thinking about «Things in the Making»* (New York: Princeton Architectural Press, 2000). See also Pauline Lefebvre, "Tracer des reprises du pragmatisme en architecture (1990-2010). Penser l'engagement des architectes avec le réel" (doctoral dissertation presented at the Université Libre de Bruxelles, 2016) as well as Ockman's own account of this endeavor elsewhere in this volume.

